# Yaw Adu-Gyamfi, PhD

Department of Civil and Environmental Engineering
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## Education

- **University of Delaware** — Newark, Delaware
  *Doctor of Philosophy* — *August 2010 - May 2013*
- **University of Delaware** — Newark, Delaware
  *Master of Civil Engineering* — *August 2008 - May 2010*
- **Kwame Nkrumah University of Science and Technology** — Kumasi, Ghana
  *Bachelor of Science in Geodetic Engineering* — *August 2003 - May 2007*

## Academic Appointments

- **Associate Professor**: University of Missouri - Columbia — September 2023 - Current
- **Assistant Professor**: University of Missouri - Columbia — August 2017 - September 2023
- **Research Scientist**: University of Virginia — April 2016 - July 2017
- **Research Associate** : Iowa State University — September 2014 - April 2016
- **Research Associate**: University of Virginia — August 2013 - September 2014

## Professional Activities

- **Associate Editor** — ASCE Journal of Transportation Engineering Part A — 2022 - Current
- **Editorial Board Member** — Road Materials and Pavement Design — 2022 - Current
- **Associate Editor** — Journal of Big Data in Transportation — 2020 - 2021
- **Guest Editor** — Special Issue of Journal of Big Data in Transportation — 2019 - 2020
- **Program Committee Member** — Data Science for Pavements Symposium — 2021 - 2022

# RESEARCH

## Refereed Journal Publications - (Student Names Underlined)

J-36 Mark Amo-Boateng, **Yaw Adu-Gyamfi**(2025). "Generative Adversarial Network for Real-time Identification and Pixel-Level Annotation of Highway Pavement Distresses", Journal of Automation in Construction, **Accepted**
Impact Factor: 9.6

J-35 Linlin Zhang, Praveen Arachchilage, Xiang Yu, **Yaw Adu-Gyamfi**(2025). "A Novel, Low-Cost, Double U-Net Model for Predicting Traffic Sign Reflectivity Intensity from Camera Data", Transportation Research Record, **Accepted**
Impact Factor: 1.9

J-34 Ashkan Behzadian, Linlin Zhang, **William Buttlar**, **Yaw Adu-Gyamfi**(2025). "Feasibility and Reliability Assessment of Inexpensive Solid-State LiDAR for Rutting Measurement in Asphalt Pavement", Transportation Research Record, **In Press**
Impact Factor: 1.9

J-33 Tanner Muturi, **Yaw Adu-Gyamfi**(2025). "Enhanced Crack Segmentation Using META's Segment Anything Model with Low-Cost Ground Truths and Multimodal Prompts", Transportation Research Record, **In Press**
Impact Factor: 1.9

J-32 Tanner Muturi, David Kesse, **Yaw Adu-Gyamfi**(2025). "Complex Shadow Removal in Pavement Imagery: Leveraging Diffusion Models for Advanced Solutions", Transportation Research Record, **In Press**
Impact Factor: 1.9

J-31 Linlin Zhang, Xiang Yu, Armstrong Aboah, **Yaw Adu-Gyamfi**(2025). "Three-Dimensional Object Detection and High-Resolution Traffic Parameter Extraction Using Low-Resolution LiDAR Data", ASCE Journal of Transportation Research Part A., Vol. 151, Iss. 3
Impact Factor: 2.2

J-30 Elizabeth Arthur, Tanner Muturi, **Yaw Adu-Gyamfi**(2024). "Training Vehicle Detection and Classification Models with Less Data: An Active Learning Approach", Transportation Research Record, Vol. 2678, Is. 11
Impact Factor: 1.9

J-29 Kojo Konadu Adu-Gyamfi, Karo Ahmadi-Dehrashid, **Yaw Adu-Gyamfi**, Pujitha Gunaratne, Anuj Sharma (2023). "MobiScout: A scalable cloud-based driving and activity monitoring platform featuring an IOS app and a WatchOS extension", SoftwareX, Vol 24, 101588, https://doi.org/10.1016/j.softx.2023.101588.
Impact Factor: 4.0
Percent Contribution: 20 percent

J-28. Linlin Zhang, Xian Yu, **Adu-Gyamfi Y.O.**, Sun Carlos, (2023), "Spatio-Temporal Fusion of LiDAR and Camera Data for Omnidirectional Depth Perception", Transportation Research Record, 0(0).
doi:https://doi.org/10.1177/03611981231184187.
Impact Factor: 1.9

J-27. Wang, Jiyang, Chai, Weiheng, Venkatachalapathy, Archana, Tan, Kai Liang, Haghighat, Arya, Velipasalar, Senem, **Adu-Gyamfi, Y.O**.; Sharma, Anuj, (2023) "Driver Head Pose Detection From Naturalistic Driving Data", IEEE Transactions on Intelligent Transportation Systems. doi: 10.1109/TITS.2021.3126231.Vol. 24, No. 9.
Impact Factor: 6.49
Percent Contribution: 15 percent

J-26. Armstrong Aboah, **Yaw Adu-Gyamfi** (2023), "Driver Maneuver Detection and Analysis using Time Series Segmentation and Classification", ASCE Journal of Transportation Research Part A., Vol. 149, No. 3.
Impact Factor: 2.18

J-25. Abdul Musah, **Yaw Adu-Gyamfi** (2022), "Machine Learning Framework for Real-Time Assessment of Traffic Safety Utilizing Connected Vehicle Data", Journal of Sustainability, Vol 14, No. 22.
doi:https://doi.org/10.3390/su142215348
Impact Factor: 3.9

J-24. Armstrong Aboah, **Yaw Adu-Gyamfi** (2022), "Mobile Sensing for Multipurpose Applications in Transportation", , Journal of Big Data Analytics in Transportation, Vol 4, pages 171–183
Impact Factor: N/A

J-23. Maged Shoman, Amoh-Boateng Mark, Adu-Gyamfi, Y.O. (2022) "Multi-Purpose, Multi-Step Deep Learning Framework for Network-Level Traffic Flow Prediction", Advances in Data Science and Adaptive Analysis, Vol. 14, No. 03, DOI: https://doi.org/10.1142/S2424922X22500103
Impact Factor: 1.3

J-22. Wang, Jiyang, Chai, Weiheng, Venkatachalapathy, Archana, Tan, Kai Liang, Haghighat, Arya, Velipasalar, Senem, **Adu-Gyamfi, Y.O**.; Sharma, Anuj, (2021) "A Survey on Driver Behavior Analysis from In-Vehicle Cameras", IEEE Transactions on Intelligent Transportation Systems. doi: 10.1109/TITS.2021.3126231.
Impact Factor: 6.49
Percent Contribution: 15 percent

J-21. Lan Phuong Uong, Vishal Mandal, Xiaofan Shu, **Adu-Gyamfi Y.O.**, Mo Zhao, (2021), "Machine Learning Framework for Improving the Accuracy of Probe Speed Data",
ASCE-SME Journal of Risk and Uncertainty in Engineering Systems, Part A. Volume 7, Issue 2.
Impact Factor: 1.59

J-20. Maged Shoman,Armstrong Aboah, **Adu-Gyamfi Y.O.**, (2020), "Deep Learning Framework for Predicting Bus Delays on Multiple Routes Using Heterogenous Datasets", Journal of Big Data Analytics in Transportation, Vol 2. Pg. 275-290.
Impact Factor: N/A

J-19. Sun, C., Feng, H., and **Adu-Gyamfi, Y.** (2020). "Characterization of Mobile Work Zone Safety in Missouri." The Open Transportation Journal, Vol. 14. Bentham Open. DOI: 10.2174/1874447802014010067.
Impact Factor: 0.71

J-18. Vishal Mandal, **Adu-Gyamfi Y.O.** (2020), "Object Detection and Tracking Algorithms for Vehicle Counting: A Comparative Analysis", Journal of Big Data Analytics in Transportation, Issue 2, pg. 251-261, https://doi.org/10.1007/s42421-020-00025-w.
Impact Factor: N/A

J-17. Vishal Mandal, Abdul Rashid Mussah, Peng Jin, **Adu-Gyamfi Y.O.** (2020), "Artificial Intelligence-Enabled Traffic Monitoring System", Journal of Sustainability, Special issue on Applications of Artificial Intelligence Based Methods in Transportation Engineering, Vol 12, Issue 21.
Impact Factor: 2.6

J-16. Aboah Armstrong, **Adu-Gyamfi Y.O.**, (2020), "Smartphone-Based Pavement Roughness Estimation Using Deep Learning with Entity Embeddings", Journal of Advances in Data Science and Adaptive Analysis. https://doi.org/10.1142/S2424922X20500072.
Impact Factor: 1.3

J-15. Farzaneh Azadi, **Yaw Adu-Gyamfi**, Carlos Sun, Praveen Edara (2020), "Work Zone Activity Real-Time Data Collection: Mobile Application Development and Testing",
Transportation Research Record: Journal of Transportation Research Board, Vol 2674, Issue 6.
Impact Factor: 1.7

J-14. Hamed Majidifard, **Yaw Adu-Gyamfi**, William G. Buttlar (2020), "Deep Machine Learning to Develop a New Asphalt Pavement Condition Index", Journal of Construction and Building Materials, Volume 247.
Impact Factor: 6.14

J-13. Hamed Majidifard, Peng Jin, **Yaw Adu-Gyamfi**, William G. Buttlar (2020), "PID: A New Benchmark Dataset to Classify and Densify Pavement Distresses", Transportation Research Record:
Journal of Transportation Research Board. doi: 10.1177/0361198120907283.
Impact Factor: 1.7

J-12. **Adu-Gyamfi Y.O.**, (2019), "GPU-Enabled Visual Analytics Framework for Big Transportation Datasets", Journal of Big Data Analytics in Transportation. Vol. 2, Issue 1. Pp 1-13.
Impact Factor: N/A

J-11. Al-Gharawi, Mohanned, **Adu-Gyamfi Y.O.**, Washer Glenn (2019), "A Framework for Automated Time-Lapse Thermography Data Processing", Journal of Construction and Building Materials, Vol. 226.
Impact Factor: 6.14

J-10. Pranamesh Chakraborty, **Adu-Gyamfi Y.O.**, Anuj, Sharma, (2018), "Traffic Congestion Detection from Camera Images using Deep Convolutional Neural Networks", Transportation Research Record:
Journal of the Transportation Research Record, Vol. 2672.
Impact Factor: 1.7

J-09. **Adu-Gyamfi Y.O.**, Sampson Asare, Anuj, Sharma, (2017), "Automated Vehicle Recognition with Deep Convolutional Neural Networks", Transportation Research Record: Journal of Transportation Research Board, Vol 2645, Issue 1, 2017.
Impact Factor: 1.7

J-08. **Adu-Gyamfi Y.O.**, Anuj, Sharma, Skylar K., Neal H., Michael J. (2017), "A Framework for Evaluating the Reliability of Wide Area Probe Data", Transportation Research Record:
Journal of Transportation Research Board, Vol 2643, Issue 1.
Impact Factor: 1.7

J-07. **Adu-Gyamfi Y.O.**, N.O. Attoh-Okine, (2014), "Traffic Sign Recognition using Sparse Representations and Active Contour Models", Transportation Research Record: Journal of the Transportation Research Board, No. 2463, pp. 35-45.
Impact Factor: 1.7

J-06. **Adu-Gyamfi Y.O.**, Titus Tienaah., N.O. Attoh-Okine, (2014), "A Functional Evaluation of Pavement Condition using a Complete Vision System", ASCE Journal of Transportation Engineering., 140(9).
Impact Factor: 1.7

J-05. **Adu-Gyamfi, Y.O.**, Carrillo Rafael, Attoh-Okine, N.O. (2012), "Multi-resolution Information Mining for Pavement Crack Image Analysis". ASCE Journal of Computing, 26(6), pp. 741-749.
Impact Factor: 4.65

J-04. **Adu-Gyamfi Y.O.)**, Attoh-Okine N.O., A.Y. Ayenu-Prah. (2010), "Critical Analysis of Different Hilbert Huang Algorithms for Pavement Profile Evaluation." ASCE Journal of Computing in Civil Engineering., 24 (6), pp.514-524.
Impact Factor: 4.65

J-03. Nii Attoh-Okine, **Adu-Gyamfi, Y.O.**, and Stephen Mensah. (2010), "Potential Application of Hybrid Belief Functions and Hilbert-Huang Transform in Layered Sensing", IEEE Sensors Journal, Vol. 11, No. 3, March 2011.
Impact Factor: 3.330

J-02. **Adu-Gyamfi Y.O.**, Attoh-Okine N.O., A.Y. Ayenu-Prah. (2009), "Pavement profile analysis using ensemble empirical mode decomposition", International Journal of Vehicle Systems Modeling and Testing - Vol. 4, No.4, pp. 277 - 287.
Impact Factor: 0.174

J-01. **Adu-Gyamfi Y.O.)**, Attoh-Okine N.O., A.Y. Ayenu-Prah. (2010), "Critical Analysis of Different Hilbert Huang Algorithms for Pavement Profile Evaluation." ASCE Journal of Computing in Civil Engineering., 24 (6), pp.514-524.
Impact Factor: 4.65

## Journal Papers Pending and Under Review

J-1. Linlin Zhang, Xiang Yu, **Yaw Adu-Gyamfi**, Anuj Sharma (2022), "Spatio - Temporal Fusion of LiDAR and Camera Data for Omnidirectional Depth Perception. ", Transportation Research Record

J-2. Mark Amo-Boating, **Yaw Adu-Gyamfi** (2022), "Accelerating Statewide Connected Vehicles Big (Sensor Fusion) Data ETL Pipelines on GPUs", ASCE Journal of Computing in Civil Engineering.

## Refereed Conference Presentations - (Student Names Underlined)

C-18. Abdul Rashid, **Yaw Adu-Gyamfi** (2022), "A Framework for Real-time Assessment of Traffic Safety Utilizing Connected Vehicle Data", Transportation Research Board.- **Under Review**

C-17. Jamal Saied Walker, **Yaw Adu-Gyamfi** (2022), "Cloud Databases for Transportation Data Storage, Retrieval and Analytics - A Comparative Analysis", Transportation Research Board - **Under Review**

C-16. Armstrong Aboah,Maged Shoman, Alex Morehead, Ye Duan, Abdulateef Daud **Yaw Adu-Gyamfi** (2022), "A Region-Based Deep Learning Approach to Automated Retail Checkout", Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition – CVPR, Pg. 3209-3214
Impact Factor: 45.17

C-15. Armstrong Aboah, **Yaw Adu-Gyamfi** (2021), "A vision-based system for traffic anomaly detection using deep learning and decision trees", Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition – CVPR, Pg. 4207 - 4212
Impact Factor: 45.17

C-14. Xiaofan Shu, **Yaw Adu-Gyamfi**, Carlos Sun, Praveen Edara (2021), "Interactive, Web-based Platform for Big Transportation Data Integration and Analytics", Transportation Research Board, Washing DC, January 11-15, 2021.

C-13. Vishal, Mandal, Abdul Rashid M., **Adu-Gyamfi, Y.O.** (2020), "Deep Learning Frameworks for Pavement Distress Classification: A Comparative Analysis", IEEE Conference on Big Data, Virtual Conference, December 10-13, 2020.
Acceptance Rate: 15 percent

C-12. Peng Jin, Xiaofan Shu, Vishal Mandal, **Yaw Adu-Gyamfi** (2020), "Detecting Traffic Anomalies Using a Vision-Based System", Transportation Research Board, Washing DC, January 11-15, 2020

C-11. Neetu Choubey, **Adu-Gyamfi Y.O.** (2019), "Automatic Conflation of Different Transportation Layers", Transportation Research Board, Washington DC, January 11 - 15, 2020.

C-10. Vishal, Mandal, Uong, Lan, **Adu-Gyamfi Y.O.**(2019), "Traffic Congestion Detection with Mask-RCNN", Transportation Research Board, Washington DC, January 13 - 17, 2019.

C-9. Vishal, Mandal, **Adu-Gyamfi Y.O.** (2018), "Deep Convolutional Neural Networks for Pavement Distress Analysis", IEEE Conference on Big Data, Seattle, WA, December 10-14, 2018.

C-8. **Adu-Gyamfi Y.O.**, Mo, Zhao (2018), "Traffic Speed Prediction for Urban Arterial Roads Using Deep Neural Networks", ASCE ICTD Conference, Pittsburgh, Pennsylvania, July 15-18, 2018.

C-7. **Adu-Gyamfi Y.O.**, Anuj Sharma (2016), "A Comprehensive Data Driven Evaluation of Wide Area Probe Data: Opportunities and Challenges", Transportation Research Board, Washington DC, January 11 - 15, 2016

C-6. **Adu-Gyamfi Y.O.**, Chase Stephen (2015), "Bridge Deck Sub-Surface Defect Detection Using Time-Lapse Thermography", Transportation Research Board, Washington DC, January 11 - 15, 2015.

C-5. **Adu-Gyamfi Y.O.**, Chase Stephen (2014), "Thermoelastic Stress Analysis of Fatigue-Prone Details", The American Society of Nondestructive Evaluation, Washington DC, August 25-27, 2014.

C-4. **Adu-Gyamfi Y.O., N.O.** Attoh-Okine, (2014), "Traffic Sign Recognition Using Sparse Representations and Active Contour Models", Transportation Research Board, Washington DC, January 13-17, 2014.

C-3. S. Rodrigues Filho, **Adu-Gyamfi Y.O.**, Adarkwa, O. A., Rodrigues, R.M., Attoh-Okine, N.O. (2014), "Airport Asphalt Pavement Profile Analysis - Ensemble Empirical Mode Decomposition Approach", Transportation Research Board, January, Washington DC, January 13-17, 2014.

C-2. **Y.O. Adu-Gyamfi**, Chandra Kambhamettu, and Nii Attoh Okine (2013), "Performance Assessment of Flexible Pavements Using Active Contour Models", Airfield and Highway Pavement Conference, Los Angeles, California, June 9-12, 2013

C-1. Sampson Asare., **Adu-Gyamfi Y.O.**, N.O. Attoh-Okine, (2013), "Freeway Incident Detection Using Hilbert Huang Transform", Transportation Research Board, Washington DC, January 13-17, 2013.

## Funded Research Grants (TOTAL: $8,156,194, PI - $4,509,859)

G-24. Sponsor: Missouri Department of Transportation
Title: "Establishment of a Public-Private Transportation Data Exchange Center"
Amount:$1,200,000
Role: **PI**
Duration: 2025 - 2026
Shared Credit: 70%

G-23. Sponsor: Missouri Department of Transportation
Title: "Accurate Collection and Reporting of Worker Presence in Work Zones"
Amount:$300,000
Role: **CO-PI**
Duration: 2025 - 2026
Shared Credit: 30%

G-22. Sponsor: Missouri Department of Transportation
Title: "Workzone Speed Study"
Amount:$155,170
Role: **CO-PI**
Duration: 2025 - 2026
Shared Credit: 20%

G-21. Sponsor: Missouri Department of Transportation
Title: "TITAN - Phase 3"
Amount:$778,640
Role: **PI**
Duration: 2025 - 2028
Shared Credit: 70%

G-20. Sponsor: Missouri Department of Transportation
Title: "Comprehensive Stream Management and Analytics System"
Amount:$69,533
Role: **PI**
Duration: 2024 - 2025
Shared Credit: 100%

G-19. Sponsor: Missouri Department of Transportation
Title: "AI-Enabled Traffic Intersection Analytics"
Amount:$89,533
Role: **PI**
Duration: 2024 - 2025
Shared Credit: 100%

G-18. Sponsor: Missouri Department of Transportation
Title: "Automated TMA Warning System"
Amount:$396,533
Role: **PI**
Collaborators: Carlos Sun **CO-PI**
Duration: 2023 - 2025
Shared Credit: 60%

G-17. Sponsor: Illinois Tollway
Title: "Utilization of FTIR for Material Characterization and Forensic Analysis "
Amount:$250,000
Role: **CO-PI**
Collaborators: William Buttlar **(PI)**
Duration: 2022 - 2023
Shared Credit: 35%

G-16. Sponsor: TOYOTA
Title: "A Scalable Cloud-Edge Enabled Driving and Activity Monitoring Platform: A Pilot Study "
Amount:$100,000 ($15,000 - MU Share)
Role: **MU PI**

Lead Institution: Iowa State University
Duration: 2021 - 2022
Shared Credit: 100%

G-15. Sponsor: National Science Foundation
Title: "CAREER: Interactive and Explainable AI for Next-Gen Transportation Systems Management"
Amount:$500,187
Role: **PI**
Duration: 2021 - 2026
Shared Credit: 100%

G-14. Sponsor: US Department of Transportation
Title: "Preparing Local Governments for Automated Driving Systems – Implementing WZDx in St. Charles County, Missouri"
Amount:$100,000
Role: **PI**
Collaborators: Praveen Edara, Carlos Sun **CO-PI**
Duration: 2021 - 2022
Shared Credit: 60%

G-13. Sponsor: City of Kansas City Missouri
Title: "Modernization of KCMOs Asphalt Pavement Evaluation and Maintenance Programs using Machine Learning and Innovative Materials"
Amount:$59,900
Role: **PI**
Collaborators: William Buttlar **CO-PI**
Duration: 2020 - 2022
Shared Credit: 60%

G-12. Sponsor: Missouri Department of Transportation
Title: "TITAN - An Interactive Web-based Platform for Transportation Data InTegration and Analytics – Phase II"
Amount:$449,932
Role: **PI**
Collaborators: Praveen Edara, Carlos Sun **CO-PI**
Duration: 2021 - 2025
Shared Credit: 50%

G-11. Sponsor: Missouri Department of Transportation
Title: "Development of a Surface Transportation Impact Factor for Winter Severity Indices"
Amount:$150,000 ($39,689 – MU Share)
Role: **MU PI**
Collaborators: Praveen Edara, Sabreena Anowar **CO-PI**
Lead Institution: Wood PLC
Duration: 2020 - 2021
Shared Credit: 40%

G-10. Sponsor: National Science Foundation CMMI
Title: "SCC-PG: Developing Food Oases in the Urban Food Deserts with Smart Urban Grower Network"
Amount:$58, 038
Role: **Senior Personnel**
Collaborators: Carlos Sun **CO-PI**
Lead Institution: University of Missouri - Kansas City
Duration: 2020 - 2021
Shared Credit: 33%

G-9. Sponsor: St. Charles County Government
Title: "DASH: An Integrated Warehouse for Data Sharing and Analytics"
Amount:$1,500,000
Role: **PI**
Collaborators: Praveen Edara, Carlos Sun, Prasad Calyam, Henry Brown **CO-PI**
Duration: 2020 - 2023
Shared Credit: 40%

G-8. Sponsor: Federal Highway Administration
Title: "DeepInsight:Deep Extraction of Driver State from Naturalistic Driving Dataset"
Amount:$1,050,000 ($149,099 – MU share)
Role: **MU PI**
Lead Institution: Iowa State University
Duration: 2020 - 2023
Shared Credit: 100%

G-7. Sponsor: Federal Highway Administration
Title: "MIMIC – Multidisciplinary Initiative on Methods to Integrate and Create Artificial Realistic Data"
Amount:$1,073,255
Role: **CO-PI**
Collaborators: Praveen Edara **PI**, Carlos Sun
Duration: 2020 - 2022
Shared Credit: 11%

G-6. Sponsor: Missouri Department of Transportation
Title: "Highway Safety Manual Training"
Amount:$100,001
Role: **CO-PI**
Collaborators: Praveen Edara, Carlos Sun **PI**
Duration: 2018 - 2019
Shared Credit: 33%

G-5. Sponsor: SWZDI TPF
Title: "Investigation of Autonomous-Connected Vehicles in Work Zones"
Amount:$74,746
Role: **CO-PI**
Collaborators: Praveen Edara, Carlos Sun **PI**
Duration: 2018 - 2019
Shared Credit: 33%

G-4. Sponsor: Wisconson Department of Transportation
Title: "Transportation Engineering & Programming Study"
Amount:$109,999
Role: **CO-PI**
Collaborators: Charles Nemmers **PI**, Carlos Sun, Praveen Edara
Duration: 2018 - 2019
Shared Credit: 20%

G-3. Sponsor: Missouri Department of Tranportation
Title: "TITAN - An Interactive Web-based Platform for Transportation Data InTegration and ANalytics"
Amount:$24,992
Role: **CO-PI**
Collaborators: Praveen Edara, Carlos Sun **PI**
Duration: 2018 - 2019
Shared Credit: 33%

G-2. Sponsor: USDOT Transportation Center
Title: "An Interactive Web-based Platform for Freight Data Integration and Analysis"
Amount:$20,001
Role: **CO-PI**
Collaborators: Praveen Edara **PI**
Duration: 2018 - 2019
Shared Credit: 33%

G-1. Sponsor: SWZDI TPF
Title: "Smart Work Zone Activity APP"
Amount:$57,133
Role: **PI**
Collaborators: Praveen Edara, Carlos Sun **CO-PI**
Duration: 2018 - 2019
Shared Credit: 40%

## Reports

R-1. Carlos, Sun, **Yaw Adu-Gyamfi**, Edara, Praveen, (2019), "TITAN – An Interactive Web-Based Platform for Transportation Data Integration and Analytics", Missouri Department of Transportation"

R-2. **Yaw Adu-Gyamfi**,Carlos, Sun, Edara, Praveen(2019), "SWIZAPP – Smart Work-Zone Activity App", Smart Work zone Pooled-Fund Study"

R-3. Chase, S.B., **Yaw Adu-Gyamfi**, Aktan, A.E., and Minaie, E.(2016), "Synthesis of National and International Methodologies Used for Bridge Health Indices",
Long-Term Bridge Performance Program, Publication No. FHWA-HRT-15-081

## STUDENT ADVISING

| Year | Undergraduate Students | Masters Students | Doctoral Students |
|---|---|---|---|
| 2017 | 0 | 0 | 0 |
| 2018 | 2 | 2 | 1 |
| 2019 | 1 | 4 | 3 |
| 2020 | 1 | 3 | 3 |
| 2021 | 4 | 1 | 6 |
| 2022 | 2 | 2 | 6 |
| 2023 | 2 | 3 | 10 |
| 2024 | 2 | 4 | 10 |
| 2025 | 2 | 3 | 8 |

### Doctoral Students Under Supervision

- Name: **Mohanned Algaraway**
  Dissertation Title: "Development of Automated Algorithms for Subsurface Defect Detection and Characterization "
  Role: CO-CHAIR
  Status: GRADUATED - SPRING 2021

- Name: **Armstrong Aboah**
  Dissertation Title: "AI-Based Framework for Automatically Extracting High-Low Features from NDS Data to Understand Drivers' Behavior"
  Role: CHAIR
  Status: GRADUATED - SPRING 2023

- Name: **Maged Shorman**
  Dissertation Title: "Interactive, Multi-purpose AI-Platform for Predictive Analytics of Traffic Flow Variables "
  Role: CHAIR
  Status: GRADUATED - FALL 2023

- Name: **Abdul Rashid**
  Dissertation Title: "Big-Data Enabled Framework for Real Time Predictions of Traffic Incidents"
  Role: CHAIR
  Status: GRADUATED - SPRING 2024

- Name: **Linlin Zhang**
  Dissertation Title: "Large Scale Traffic Analytics via Edge Computing and Computer Vision "
  Role: CHAIR
  Status: GRADUATED - FALL 2024

- Name: **Neema Jakisa Owor**
  Dissertation Title: "Generative AI Applications in Transportation"
  Role: CHAIR
  Status: Expected - FALL 2025

- Name: **Elizabeth Arthur**
  Dissertation Title: "Leveraging AI for Next Generation Traffic Management System"
  Role: CHAIR
  Status: Expected - FALL 2025

- Name: **Tanner Muturi**
  Dissertation Title: "Sidewalk Surveillance - Robotic Development"
  Role: CHAIR
  Status: Expected - FALL 2026

- Name: **Lan Uong**
  Dissertation Title: "Traffic Anomaly Detection using Big Data"
  Role: CHAIR
  Status: Expected - FALL 2026

- Name: **Xian Yu**
  Dissertation Title: "Design and Implementation of a Low-Cost, Autonomous TMA"
  Role: CHAIR
  Status: Expected - FALL 2026

## Masters Students Under Supervision

- Name: **Huang Feng**
  Dissertation Title: "TMA work zone safety analysis and modelling using machine learning"
  Role: CO-CHAIR
  Status: Graduated - SPRING 2019

- Name: **Vishal Mandal**
  Dissertation Title: "Artificial intelligence enabled automatic traffic monitoring system"
  Role: CHAIR
  Status: Graduated - FALL 2019

- Name: **Neetu Choubey**
  Dissertation Title: "Automated conflation framework for integrating transportation big datasets"
  Role: CHAIR
  Status: Graduated - FALL 2019

- Name: **Jin Peng**
  Dissertation Title: "Traffic Anomaly detection with deep learning"
  Role: CHAIR
  Status: Graduated - SPRING 2020

- Name: **Xiaofan Shu**
  Dissertation Title: "TITAN: an interactive, web-based platform for transportation data integration and analytics"
  Role: CHAIR
  Status: Graduated - SPRING 2020

- Name: **Mohammadmehdi Zoghifard**
  Dissertation Title: " Truck-Mounted Attenuator Alert System - Development and Testing"
  Role: CHAIR
  Status: Graduated - SUMMER 2022

- Name: **Charles Townsend**
  Dissertation Title: "Leveraging Artificial Intelligence to detect inequities in pavement rehabilitation in Missouri"
  Role: CHAIR
  Status: Graduated - SPRING 2022

- Name: **Daud Abdulateef**
  Dissertation Title: "Edge Computing-Enabled Road Condition Monitoring: System Development and Evaluation"
  Role: CHAIR
  Status: Graduated - SPRING 2023

- Name: **Ahnaf Zahin**
  Dissertation Title: "Multi-head Self-Attention Mechanism-based Transformer Model For Predicting Bus Arrival Times Across Multiple Bus Routes Using Heterogenous Data"
  Role: CHAIR
  Status: Graduated - SPRING 2023

- Name: **Jacob Dodge**
  Dissertation Title: "Sidewalk Condition Index - A novel Approach"
  Role: CHAIR
  Status: GRADUATED - FALL 2024

- Name: **Abby Watts**
  Dissertation Title: "Traffic Sign Condition Index - A novel Approach"
  Role: CHAIR
  Status: Expected - SPRING 2025

- Name: **Kang Gyeongmin**
  Dissertation Title: "Audible TMA Alerts - Simulator Study"
  Role: CHAIR
  Status: Expected - SPRING 2025

## Postdoctoral Fellow Training and Fellowship

- **Mark Amoh-Boateng, Phd** 2021 - current
  Degree Institution: Hohai University, China

## Thesis Committee Involvement

- Name: **John Patman**
  Thesis Title: "Intelligence-driven edge computing for visual cloud computing systems"
  Degree: MS
  Major Department: Computer Science
  Status: Graduated - SPRING 2018

- Name: **Srivastava Tanya**
  Thesis Title: "Simulation of spiral-shaped mems human energy harvester using piezoelectric transduction"
  Degree: MS
  Major Department: Electrical Engineering
  Status: Graduated - SPRING 2018

- Name: **Antequera Ronny Bazan**
  Thesis Title: "Custom templates based heterogenous resource allocation for data-intensive applications"
  Degree: PHD
  Major Department: Computer Science
  Status: Graduated - SPRING 2020

- Name: **Zhang Siyang**
  Thesis Title: "Investigation of smart work zone technologies using mixed simulator and field studies"
  Degree: PHD
  Major Department: Civil and Environmental Engineering
  Status: Graduated - SPRING 2020

- Name: **Neely Travis**
  Thesis Title: "Automated defense against attacks using suspiciousness tracking"
  Degree: MS
  Major Department: Computer Science
  Status: Graduated - SPRING 2021

- Name: **Majidifard Hamed**
  Thesis Title: "Developing a modern recycled mix design and automatic pavement monitoring system"
  Degree: PHD
  Major Department: Civil and Environmental Engineering
  Status: Graduated - SPRING 2021

- Name: **Fahad Alsharari**
  Thesis Title: "Fatigue and static behavior of post-tensioned steel- concrete composite beams"
  Degree: PHD
  Major Department: Civil and Environmental Engineering
  Status: Graduated - SPRING 2021

- Name: **Osunkoya, Opeoluwa**
  Thesis Title: "Function centric computing for computer vision applications"
  Degree: MS
  Major Department: Computer Science
  Status: Graduated - SPRING 2021

- Name: **Haimanti, Bala**
  Thesis Title: "Understanding acceptance of autonomous mobility services using statistical and deep learning approaches"
  Degree: MS
  Major Department: Civil and Environmental Engineering
  Status: Graduated - FALL 2021

- Name: **Vaibhav V Akashe**
  Thesis Title: "Securing cloud-hosted applications using active defense with rule-based adaptations"
  Degree: MS
  Major Department: Computer Science
  Status: Graduated - FALL 2021

## COURSES TAUGHT

### Undergraduate

- Course Number: CVENG 3100
  Course Title: Fundamentals of Transportation Engineering
  Number of Credits: 4
  Terms Offered: 1
  Enrollment: 45
  Overall Course Evaluation Score: 4.42/5

- Course Number: ENGR 1200
  Course Title: Statics and Elementary Strength of Materials
  Number of Credits: 3
  Terms Offered: 8
  Enrollment: 200
  Overall Course Evaluation Score: 4.30/5

- Course Number: CVENG 4001
  Course Title: Urban Transportation Data Science
  Number of Credits: 3
  Terms Offered: 2
  Enrollment: 4
  Overall Course Evaluation Score: 4.27/5

### Graduate

- Course Number: CVENG 8160
  Course Title: Advanced Research Methods in Civil Engineering
  Number of Credits: 3
  Terms Offered: 1
  Enrollment: 8
  Overall Course Evaluation Score: 4.72/5

- Course Number: CVENG 7001
  Course Title: Urban Transportation Data Science
  Number of Credits: 3
  Terms Offered: 2
  Enrollment: 4
  Overall Course Evaluation Score: 4.27/5

## SERVICE

### College Service

- **Dean Search Committee Member** 2020
- **CEE Representative: Annual Lectureship Standing Committee** 2018 - Current

### National Committee Involvement

- **Member:** Artificial Intelligence and Advanced Computing Applications, TRB AED50 2021 - Current
- **Associate Editor:** ASCE Journal of Transportation Engineering - Part A 2021 - Current
- **Member:** ITE Diversity Scholarship Committee 2020 - Current

### Journal Reviews

- ASCE-ASME Journal of Risk and Uncertainty in Engineering Systems. 2019 - Current
- ADSAA Advances in Data Science and Adaptive Analysis 2014 - Current
- TRB – Artificial Intelligence Committee. 2013 - Current
- IEEE ITS Transactions 2020 - Current
- Construction and Building Materials 2020 - Current